分類	編號	論文標題	年份	發表出處	製程節點	主題 / 關鍵字	備註	檔案路徑
01 動態比較器與 StrongARM Latch	1	The StrongARM Latch: A Circuit for All Seasons	2015	IEEE Solid-State Circuits Magazine	—	StrongARM latch、比較器經典教學、再生放大	經典理論	01_動態比較器與StrongARM_Latch/01_The StrongARM Latch - A Circuit for All Seasons (Razavi 2015).pdf
01 動態比較器與 StrongARM Latch	2	Understanding the Regenerative Comparator Circuit	2014	Abidi & Xu, UCLA	—	再生比較器、動態分析、比較器理論	經典理論	01_動態比較器與StrongARM_Latch/02_Understanding the Regenerative Comparator Circuit (Abidi 2014).pdf
01 動態比較器與 StrongARM Latch	3	A 1-V, 3-GHz Strong-Arm Latch Voltage Comparator for High Speed Applications	2020	IEEE TCAS-II	65nm CMOS	高速比較器、平行路徑、offset	架構改良	01_動態比較器與StrongARM_Latch/03_A 1-V 3-GHz StrongARM Latch Voltage Comparator (TCAS-II 2020).pdf
01 動態比較器與 StrongARM Latch	4	A 1.2-V Dynamic Bias Latch-Type Comparator in 65-nm CMOS With 0.4-mV Input Noise	2018	IEEE JSSC	65nm CMOS	動態偏壓、低雜訊、double-tail	架構改良	01_動態比較器與StrongARM_Latch/04_A 1.2-V Dynamic Bias Latch-Type Comparator with 0.4-mV Noise (JSSC 2018).pdf
01 動態比較器與 StrongARM Latch	5	A 19 fJ/op, Low-Offset StrongARM Latch Comparator for Low-Power High-Speed Applications	2024	IEEE ISCAS	TSMC 65nm	低功耗、低offset、高速比較器	架構改良	01_動態比較器與StrongARM_Latch/05_A 19 fJ-op Low-Offset StrongARM Latch Comparator (ISCAS 2024).pdf
01 動態比較器與 StrongARM Latch	6	Analysis and Design of a Low-Voltage Low-Power Double-Tail Comparator	2014	IEEE TVLSI	0.18μm CMOS	double-tail、延遲分析、低電壓	替代架構	01_動態比較器與StrongARM_Latch/06_Analysis and Design of a Low-Voltage Low-Power Double-Tail Comparator (TVLSI 2014).pdf
01 動態比較器與 StrongARM Latch	7	Design of Strong-Arm Latch Comparator for Low Power 12-bit SAR ADC	2024	IEEE ICICACS	—	SAR ADC 比較器、低功耗	系統應用	01_動態比較器與StrongARM_Latch/07_Design of Strong-Arm Latch Comparator for Low-Power 12-bit SAR ADC (ICICACS 2024).pdf
01 動態比較器與 StrongARM Latch	8	Kickback Noise Reduction Techniques for CMOS Latched Comparators	2006	IEEE TCAS-II	0.18μm CMOS	kickback 雜訊、latched comparator	特定技術問題	01_動態比較器與StrongARM_Latch/08_Kickback Noise Reduction Techniques for CMOS Latched Comparators (TCAS-II 2006).pdf
01 動態比較器與 StrongARM Latch	9	Executive Summary — 6nm FinFET StrongARM Latch 設計要點	2026	自製筆記	6nm FinFET	6nm、StrongARM、設計摘要	自製筆記（非發表論文）	01_動態比較器與StrongARM_Latch/09_[自製摘要] 6nm FinFET StrongARM Latch 設計要點 Executive Summary.pdf
01 動態比較器與 StrongARM Latch	10	High-Speed StrongARM Latch Comparator Design in 6nm FinFET: Core Techniques and Challenges	2026	自製筆記 (by Claude)	6nm FinFET	6nm、再生時間常數、offset校正	自製筆記（AI 產生）	01_動態比較器與StrongARM_Latch/10_[自製筆記] High-Speed StrongARM Latch in 6nm FinFET - Core Techniques (by Claude).pdf
01 動態比較器與 StrongARM Latch	11	StrongARM Latch Comparator High-Speed Design Techniques and Literature Review for 6nm FinFET Process	2026	自製筆記 (by Claude)	6nm FinFET	6nm、文獻回顧、double-tail、fin量化	自製筆記（AI 產生）	01_動態比較器與StrongARM_Latch/11_[自製筆記] StrongARM Latch 6nm FinFET 高速設計技術與文獻回顧 (by Claude).pdf
02 SAR ADC 架構與電路設計	1	A 10-Bit 500-MS/s Pipelined SAR ADC With Feedback Factor Compensation in 6-nm FinFET	2025	IEEE TCAS-II	6nm FinFET	pipelined SAR、負電容NC、residue amp		02_SAR_ADC架構與電路設計/01_A 10-Bit 500-MS-s Pipelined SAR ADC with Feedback Factor Compensation in 6-nm FinFET (TCAS-II 2025).pdf
02 SAR ADC 架構與電路設計	2	A 9.1-ENOB 6-mW 10-Bit 500-MS/s Pipelined-SAR ADC With Current-Mode Residue Processing in 28-nm CMOS	2019	IEEE JSSC	28nm CMOS	pipelined-SAR、電流模residue		02_SAR_ADC架構與電路設計/02_A 9.1-ENOB 10-Bit 500-MS-s Pipelined-SAR ADC with Current-Mode Residue (JSSC 2019).pdf
02 SAR ADC 架構與電路設計	3	A 12b 600MS/s Pipelined SAR and 2×-Interleaved Incremental Delta-Sigma ADC with Source-Follower-Based Residue-Transfer Scheme in 7nm FinFET	2021	ISSCC (10.5)	7nm FinFET	pipelined SAR、incremental DSM、source-follower residue		02_SAR_ADC架構與電路設計/03_A 12b 600MS-s Pipelined SAR and 2x-Interleaved Incremental Delta-Sigma ADC in 7nm FinFET (ISSCC 2021).pdf
02 SAR ADC 架構與電路設計	4	A Novel Approach to Design SAR-ADC: Design Partitioning Method	2016	IEEE TCAD	—	SAR ADC 建模、LUT、架構評估		02_SAR_ADC架構與電路設計/04_A Novel Approach to Design SAR-ADC - Design Partitioning Method (TCAD 2016).pdf
02 SAR ADC 架構與電路設計	5	Design and Implementation of High Speed and Low Power 12-bit SAR ADC using 22nm FinFET	2022	WSEAS Trans. Systems & Control	22nm FinFET	12-bit SAR、低功耗、OTA		02_SAR_ADC架構與電路設計/05_Design and Implementation of High-Speed Low-Power 12-bit SAR ADC using 22nm FinFET (WSEAS 2022).pdf
02 SAR ADC 架構與電路設計	6	Design and Simulation of 10-Bit SAR ADC Using 18nm FinFET Technology	2024	IEEE ISCAIE	18nm FinFET	10-bit SAR、模擬驗證		02_SAR_ADC架構與電路設計/06_Design and Simulation of 10-Bit SAR ADC Using 18nm FinFET (ISCAIE 2024).pdf
02 SAR ADC 架構與電路設計	7	Design, Implementation, and Analysis of a Low-Power SAR ADC for Biomedical Applications	2025	碩士論文 (CSU Long Beach)	—	生醫應用、低功耗SAR、碩士論文	學位論文（84頁）	02_SAR_ADC架構與電路設計/07_Design, Implementation and Analysis of a Low-Power SAR ADC for Biomedical Applications (MSc Thesis 2025).pdf
02 SAR ADC 架構與電路設計	8	Highly Efficient Fin type Field Effect Transistor (FinFET)-Based Analog to Digital Convertor Design for High-Speed Digital Electronics Applications	2025	Cureus Journal of Engineering	FinFET	FinFET ADC、高速數位應用		02_SAR_ADC架構與電路設計/08_Highly Efficient FinFET-Based Analog-to-Digital Converter Design (Cureus 2025).pdf
03 SAR ADC 自動化佈局與生成工具	1	SAR ADC Layout Generation Using Digital Place-and-Route Tools	2023	Tsai & Liu, NTU (研討會)	—	DPR 佈局、macro-cell、寄生抑制		03_SAR_ADC自動化佈局與生成工具/01_SAR ADC Layout Generation Using Digital Place-and-Route Tools (Tsai & Liu 2023).pdf
03 SAR ADC 自動化佈局與生成工具	2	A 0.0067-mm² 12-bit 20-MS/s SAR ADC Using Digital Place-and-Route Tools in 40-nm CMOS	2022	IEEE TVLSI	40nm CMOS	DPR 佈局、SAR ADC、超小面積		03_SAR_ADC自動化佈局與生成工具/02_A 0.0067-mm2 12-bit 20-MS-s SAR ADC Using Digital Place-and-Route Tools in 40-nm CMOS (TVLSI 2022).pdf
03 SAR ADC 自動化佈局與生成工具	3	OpenSAR: An Open Source Automated End-to-end SAR ADC Compiler	2021	IEEE/ACM ICCAD	—	開源、端到端SAR編譯器、自動P&R		03_SAR_ADC自動化佈局與生成工具/03_OpenSAR - An Open Source Automated End-to-end SAR ADC Compiler (ICCAD 2021).pdf
03 SAR ADC 自動化佈局與生成工具	4	A Circuit-Design-Driven Tool with a Hybrid Automation Approach for SAR ADCs in IoT	2017	IEEE A-SSCC (研討會)	40nm CMOS	混合自動化、IoT SAR、樣板佈局	與下方期刊版為同團隊研討會前身	03_SAR_ADC自動化佈局與生成工具/04_A Circuit-Design-Driven Tool with a Hybrid Automation Approach for SAR ADCs in IoT (ASSCC 2017).pdf
03 SAR ADC 自動化佈局與生成工具	5	A Hybrid Design Automation Tool for SAR ADCs in IoT	2018	IEEE TVLSI	40nm CMOS	混合自動化、IoT SAR、規格到佈局	期刊版；4-08463579 為其重複版本	03_SAR_ADC自動化佈局與生成工具/05_A Hybrid Design Automation Tool for SAR ADCs in IoT (TVLSI 2018).pdf
03 SAR ADC 自動化佈局與生成工具	6	A 500-MS/s 8-bit SAR ADC Generated from an Automated Layout Generation Framework in 14-nm FinFET Technology	2024	研討會 (Hanyang Univ.)	14nm FinFET	自動佈局框架、FinFET SAR		03_SAR_ADC自動化佈局與生成工具/06_A 500-MS-s 8-bit SAR ADC from an Automated Layout Generation Framework in 14-nm FinFET (2024).pdf
03 SAR ADC 自動化佈局與生成工具	7	PASTEL: Parasitic Matching-Driven Placement and Routing of Capacitor Arrays With Generalized Ratios in Charge-Redistribution SAR ADCs	2020	IEEE TCAD	—	電容陣列P&R、寄生匹配、非線性最小化		03_SAR_ADC自動化佈局與生成工具/07_PASTEL - Parasitic Matching-Driven Placement and Routing of Capacitor Arrays in SAR ADCs (TCAD 2020).pdf
03 SAR ADC 自動化佈局與生成工具	8	Placement for Binary-Weighted Capacitive Array in SAR ADC Using Multiple Weighting Methods	2014	IEEE TCAD	—	電容陣列擺放、匹配、失配抑制		03_SAR_ADC自動化佈局與生成工具/08_Placement for Binary-Weighted Capacitive Array in SAR ADC Using Multiple Weighting Methods (TCAD 2014).pdf
04 AI / 機器學習輔助設計	1	A Global-Local Optimization Approach for Asynchronous SAR ADC Design	2024	Univ. of Glasgow / Georgia Tech	—	ANN代理模型、差分演化、SAR最佳化		04_AI機器學習輔助設計/01_A Global-Local Optimization Approach for Asynchronous SAR ADC Design (ANN surrogate).pdf
04 AI / 機器學習輔助設計	2	An LLM-assisted Analog IC Design Tool with Automatic Topology Selection and Circuit Sizing	2025	IEEE SMACD (NYCU)	—	LLM、拓樸選擇、ANN sizing		04_AI機器學習輔助設計/02_An LLM-assisted Analog IC Design Tool with Automatic Topology Selection and Circuit Sizing (SMACD 2025).pdf
04 AI / 機器學習輔助設計	3	Analog/Mixed-Signal Circuit Synthesis Enabled by the Advancements of Circuit Architectures and Machine Learning Algorithms	2022	ASP-DAC (USC)	—	AMS 合成、數位密集架構、ML		04_AI機器學習輔助設計/03_Analog Mixed-Signal Circuit Synthesis Enabled by Circuit Architectures and ML (ASP-DAC 2022).pdf
04 AI / 機器學習輔助設計	4	GNN-RE: Graph Neural Networks for Reverse Engineering of Gate-Level Netlists	2022	IEEE TCAD	—	GNN、網表逆向工程、子電路分類		04_AI機器學習輔助設計/04_GNN-RE - Graph Neural Networks for Reverse Engineering of Gate-Level Netlists (TCAD 2022).pdf
04 AI / 機器學習輔助設計	5	Generative AI for Analog Integrated Circuit Design: Methodologies and Applications	2025	IEEE Access	—	生成式AI、GNN/LLM/VAE、回顧	回顧文章	04_AI機器學習輔助設計/05_Generative AI for Analog Integrated Circuit Design - Methodologies and Applications (IEEE Access 2025).pdf
04 AI / 機器學習輔助設計	6	Joint Optimization of Sequential Elements and Metal Layer Usage for Advanced Semiconductor Physical Design Using Artificial Intelligence and Machine Learning	2025	JCSTS (AMD)	<2nm	GNN+強化學習、金屬層分配、物理設計		04_AI機器學習輔助設計/06_Joint Optimization of Sequential Elements and Metal Layer Usage Using AI-ML (JCSTS 2025).pdf
04 AI / 機器學習輔助設計	7	Toward End-to-End Analog Design Automation with ML and Data-Driven Approaches (Invited Paper)	2024	ASP-DAC (UT Austin)	—	端到端自動化、layout-aware sizing、回顧	邀請論文	04_AI機器學習輔助設計/07_Toward End-to-End Analog Design Automation with ML and Data-Driven Approaches (ASP-DAC 2024).pdf
04 AI / 機器學習輔助設計	8	AnalogAgent: Self-Improving Analog Circuit Design Automation with LLM Agents	2026	arXiv (NTU Singapore)	—	LLM agent、自我改進、類比自動化		04_AI機器學習輔助設計/08_AnalogAgent - Self-Improving Analog Circuit Design Automation with LLM Agents (arXiv 2026).pdf
04 AI / 機器學習輔助設計	9	LayoutCopilot: An LLM-Powered Multiagent Collaborative Framework for Interactive Analog Layout Design	2025	IEEE TCAD (PKU)	—	LLM 多代理、互動式類比佈局		04_AI機器學習輔助設計/09_LayoutCopilot - An LLM-Powered Multiagent Framework for Interactive Analog Layout Design (TCAD 2025).pdf
04 AI / 機器學習輔助設計	10	Large Language Model (LLM) for Standard Cell Layout Design Optimization	2024	IEEE LAD Workshop (NVIDIA)	2nm	LLM、標準單元佈局、PPA最佳化		04_AI機器學習輔助設計/10_Large Language Model (LLM) for Standard Cell Layout Design Optimization (LAD 2024).pdf
04 AI / 機器學習輔助設計	11	Human Language to Analog Layout Using GLayout Layout Automation Framework	2026	研討會 / arXiv	—	自然語言、GLayout、類比佈局自動化		04_AI機器學習輔助設計/11_Human Language to Analog Layout Using GLayout Layout Automation Framework (2026).pdf
04 AI / 機器學習輔助設計	12	SPICED+: Syntactical Bug Pattern Identification and Correction of Trojans in A/MS Circuits Using LLM-Enhanced Detection	2025	IEEE TVLSI	—	LLM、SPICE netlist、木馬偵測與修正		04_AI機器學習輔助設計/12_SPICED+ - Bug Pattern Identification and Correction of Trojans in A-MS Circuits Using LLM (TVLSI 2025).pdf
04 AI / 機器學習輔助設計	13	Resilient Analog Instance Language (RAIL) Enabled Open-Source Mixed-Signal Circuits	2023	AFRL 技術報告 (Univ. of Washington)	—	開源、RAIL 語言、混合訊號自動化	技術報告（177頁）	04_AI機器學習輔助設計/13_Resilient Analog Instance Language (RAIL) Enabled Open-Source Mixed-Signal Circuits (AFRL 2023).pdf
05 高速有線傳輸應用 (PAM4/Transceiver)	1	A 112-Gb/s PAM-4 Long-Reach Wireline Transceiver Using a 36-Way Time-Interleaved SAR ADC and Inverter-Based RX Analog Front-End in 7-nm FinFET	2021	IEEE JSSC	7nm FinFET	PAM4、有線收發機、時間交錯SAR、RX AFE		05_高速有線傳輸應用_PAM4/01_A 112-Gb-s PAM-4 Long-Reach Wireline Transceiver with 36-Way TI SAR ADC in 7-nm FinFET (JSSC 2021).pdf
05 高速有線傳輸應用 (PAM4/Transceiver)	2	A 56-Gb/s PAM4 Wireline Transceiver Using a 32-Way Time-Interleaved SAR ADC in 16-nm FinFET	2017	IEEE JSSC	16nm FinFET	PAM4、有線收發機、時間交錯SAR、CTLE		05_高速有線傳輸應用_PAM4/02_A 56-Gb-s PAM4 Wireline Transceiver with 32-Way TI SAR ADC in 16-nm FinFET (JSSC 2017).pdf
05 高速有線傳輸應用 (PAM4/Transceiver)	3	A 64-Gb/s 0.33-pJ/bit PAM4 Receiver Analog Front-End with a Single-Stage Triple-Peaking CTLE Achieving 22.5-dB Boost in 40-nm CMOS	2024	IEEE ICAS	40nm CMOS	PAM4 RX AFE、三峰CTLE、VGA		05_高速有線傳輸應用_PAM4/03_A 64-Gb-s 0.33-pJ-bit PAM4 Receiver Analog Front-End with Triple-Peaking CTLE in 40-nm CMOS (ICAS 2024).pdf
05 高速有線傳輸應用 (PAM4/Transceiver)	4	A 46GS/s 7-bit Time-Interleaved Time-Domain ADC with Synthesizable Unit ADCs in 16nm FinFET	2025	研討會 (Texas A&M / GIST)	16nm FinFET	時間域ADC、時間交錯、可合成單元ADC		05_高速有線傳輸應用_PAM4/04_A 46GS-s 7-bit Time-Interleaved Time-Domain ADC with Synthesizable Unit ADCs in 16nm FinFET (2025).pdf
06 先進製程元件與物理設計	1	6nm FinFET IC 佈局注意事項及參考範例	2026	自製筆記	6nm FinFET	佈局規則、多重曝光、寄生/自熱	自製筆記（非發表論文）	06_先進製程元件與物理設計/01_[自製筆記] 6nm FinFET IC 佈局注意事項及參考範例.pdf
06 先進製程元件與物理設計	2	Comparative Characterization of NWFET and FinFET Transistor Structures Using TCAD Modeling	2022	Micromachines (MDPI)	14/7nm	TCAD、NWFET vs FinFET、電熱特性		06_先進製程元件與物理設計/02_Comparative Characterization of NWFET and FinFET Transistor Structures Using TCAD Modeling (Micromachines 2022).pdf
06 先進製程元件與物理設計	3	Design Challenges and Techniques for 5nm FinFET CMOS Analog/Mixed-Signal Circuits	2023	IEEE VLSID	5nm FinFET	5nm AMS、設計挑戰、LDE		06_先進製程元件與物理設計/03_Design Challenges and Techniques for 5nm FinFET CMOS Analog-Mixed-Signal Circuits (VLSID 2023).pdf
06 先進製程元件與物理設計	4	Design Rule Evaluation Framework Using Automatic Cell Layout Generator for Design Technology Co-Optimization	2019	IEEE TVLSI	14nm 及以下	DTCO、自動單元佈局、設計規則評估		06_先進製程元件與物理設計/04_Design Rule Evaluation Framework Using Automatic Cell Layout Generator for DTCO (TVLSI 2019).pdf
06 先進製程元件與物理設計	5	Design Technology Co-optimization for RF/mmWave Circuits with Circuit under Inductor (CUI) in FinFET CMOS Technologies	2025	IEEE RFIC (TSMC)	6nm/16nm CMOS	DTCO、RF/mmWave、CUI、面積縮減		06_先進製程元件與物理設計/05_Design Technology Co-optimization for RF-mmWave Circuits with Circuit under Inductor in FinFET (RFIC 2025).pdf
06 先進製程元件與物理設計	6	Enhanced Optimal Multi-Row Detailed Placement for Neighbor Diffusion Effect Mitigation in Sub-10 nm VLSI	2019	IEEE TCAD (Kahng)	sub-10nm	細部擺放、NDE、動態規劃		06_先進製程元件與物理設計/06_Enhanced Optimal Multi-Row Detailed Placement for Neighbor Diffusion Effect Mitigation in Sub-10nm (TCAD 2019).pdf
06 先進製程元件與物理設計	7	From Interconnect Materials and Processes to Chip Level Performance: Modeling and Design for Conventional and Exploratory Concepts	2020	IEEE IEDM (Georgia Tech / Samsung)	7nm	連線縮放、銅/釕、系統級評估		06_先進製程元件與物理設計/07_From Interconnect Materials and Processes to Chip Level Performance (IEDM 2020).pdf
06 先進製程元件與物理設計	8	Hybrid Integration of Gate-All-Around Stacked Si Nanosheet FET and Si/SiGe Super-Lattice FinFET to Optimize 6T-SRAM for N3 Node and Beyond	2024	IEEE T-ED	3nm (N3)	GAA 奈米片、SL-FinFET、6T-SRAM		06_先進製程元件與物理設計/08_Hybrid Integration of GAA Nanosheet FET and SL-FinFET to Optimize 6T-SRAM for N3 Node (T-ED 2024).pdf
06 先進製程元件與物理設計	9	Interconnect Design and Technology Optimization for Conventional and Emerging Nanoscale Devices: A Physical Design Perspective	2018	IEEE IEDM (Invited)	—	連線設計、元件/連線協同最佳化、FinFET/TFET	邀請論文	06_先進製程元件與物理設計/09_Interconnect Design and Technology Optimization for Conventional and Emerging Nanoscale Devices (IEDM 2018).pdf
06 先進製程元件與物理設計	10	New structure transistors for advanced technology node CMOS ICs	2024	National Science Review	先進節點	新型電晶體、GAA、元件回顧	回顧文章	06_先進製程元件與物理設計/10_New Structure Transistors for Advanced Technology Node CMOS ICs (Natl. Science Review 2024).pdf

06 先進製程元件與物理設計	11	Optimizing Power, Performance, and Signal Integrity in Sub-6nm Physical Design Architectures	2025	IEEE GCAT	sub-6nm	物理設計、IR-drop、訊號完整性		06_先進製程元件與物理設計/11_Optimizing Power, Performance and Signal Integrity in Sub-6nm Physical Design Architectures (GCAT 2025).pdf
06 先進製程元件與物理設計	12	Physical Design Challenges and Solutions in Advanced Technology Nodes: A Comprehensive Analysis	2025	JCSTS	先進節點	物理設計、挑戰與解法、回顧		06_先進製程元件與物理設計/12_Physical Design Challenges and Solutions in Advanced Technology Nodes - A Comprehensive Analysis (JCSTS 2025).pdf
06 先進製程元件與物理設計	13	Yield and Reliability Challenges at 7nm and Below	2019	PDF Solutions / CMU (研討會)	7nm/5nm	良率、可靠度、設計規則縮放		06_先進製程元件與物理設計/13_Yield and Reliability Challenges at 7nm and Below (2019).pdf
06 先進製程元件與物理設計	14	Detailed Placement in Advanced Technology Nodes: A Survey	2016	研討會 Invited (UT Austin / CUHK)	—	細部擺放、先進節點、綜述	綜述/邀請論文	06_先進製程元件與物理設計/14_Detailed Placement in Advanced Technology Nodes - A Survey (Invited 2016).pdf
重複檔案		SAR ADC Layout Generation Using Digital Place-and-Route Tools（與 03 分類同一篇，完全重複）	2023	—	—	重複檔案	與 03 分類的同名論文完全相同	_重複檔案/[重複] SAR ADC Layout Generation Using Digital Place-and-Route Tools.pdf
重複檔案		A Hybrid Design Automation Tool for SAR ADCs in IoT（TVLSI 2018 期刊版重複，機構典藏版本）	2018	—	—	重複檔案	與 03 分類 A Hybrid Design Automation Tool 為同一篇	_重複檔案/[重複] A Hybrid Design Automation Tool for SAR ADCs in IoT (repository copy).pdf